



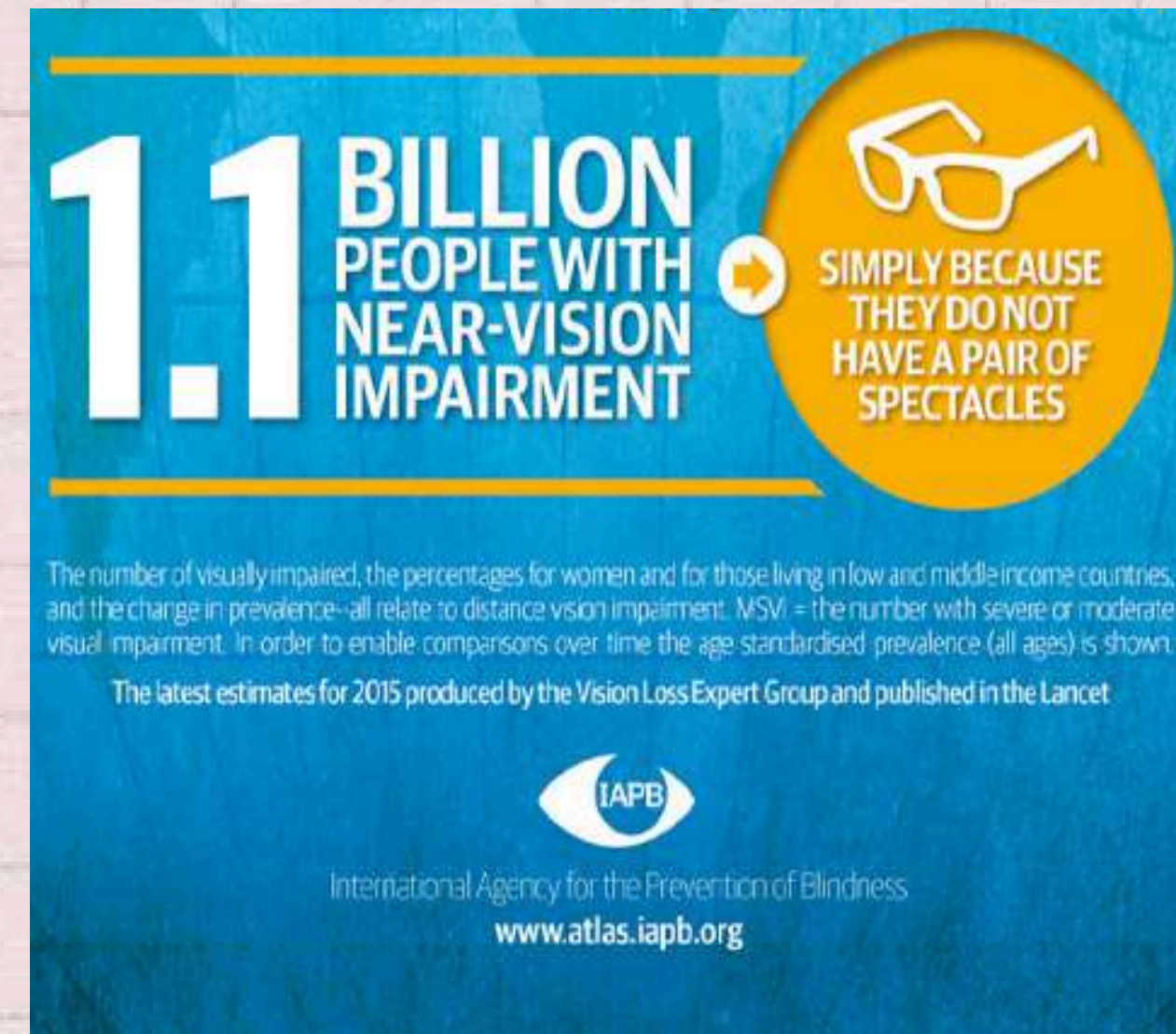
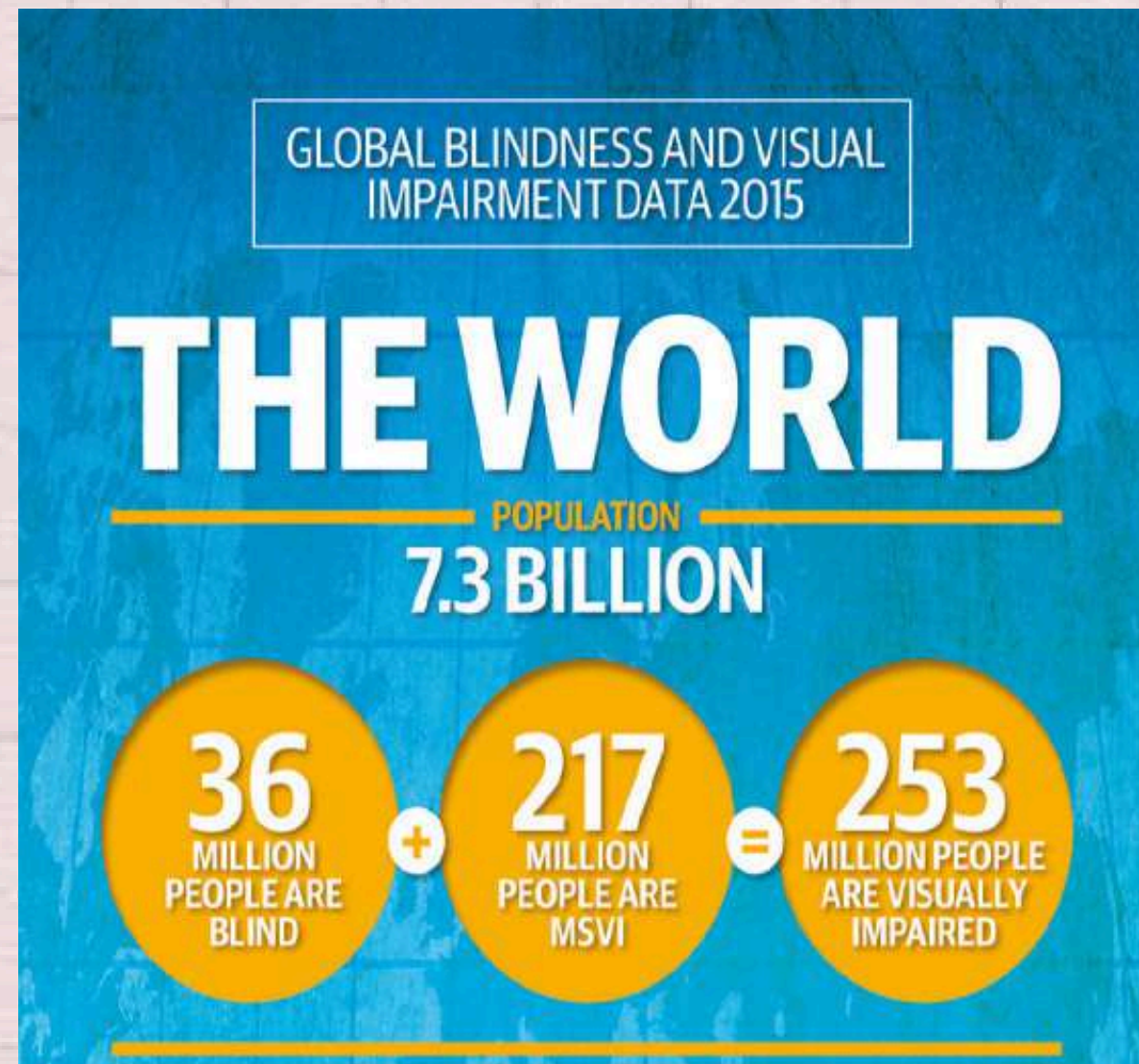
SMART

BLiND STiCK

Presentation By

ADRIENNA ARISHA (2025417326)

introduction



Statistics were given by World's Vision Loss Expert Group

iNTRODUCTION

Major challenges are being faced by visually impaired community in having a safe mobility in their daily life in both familiar events and unfamiliar events. Therefore, this community always faces injuries or death risk whenever and wherever they are.

**build a low-cost,
portable, and user-
friendly device.**

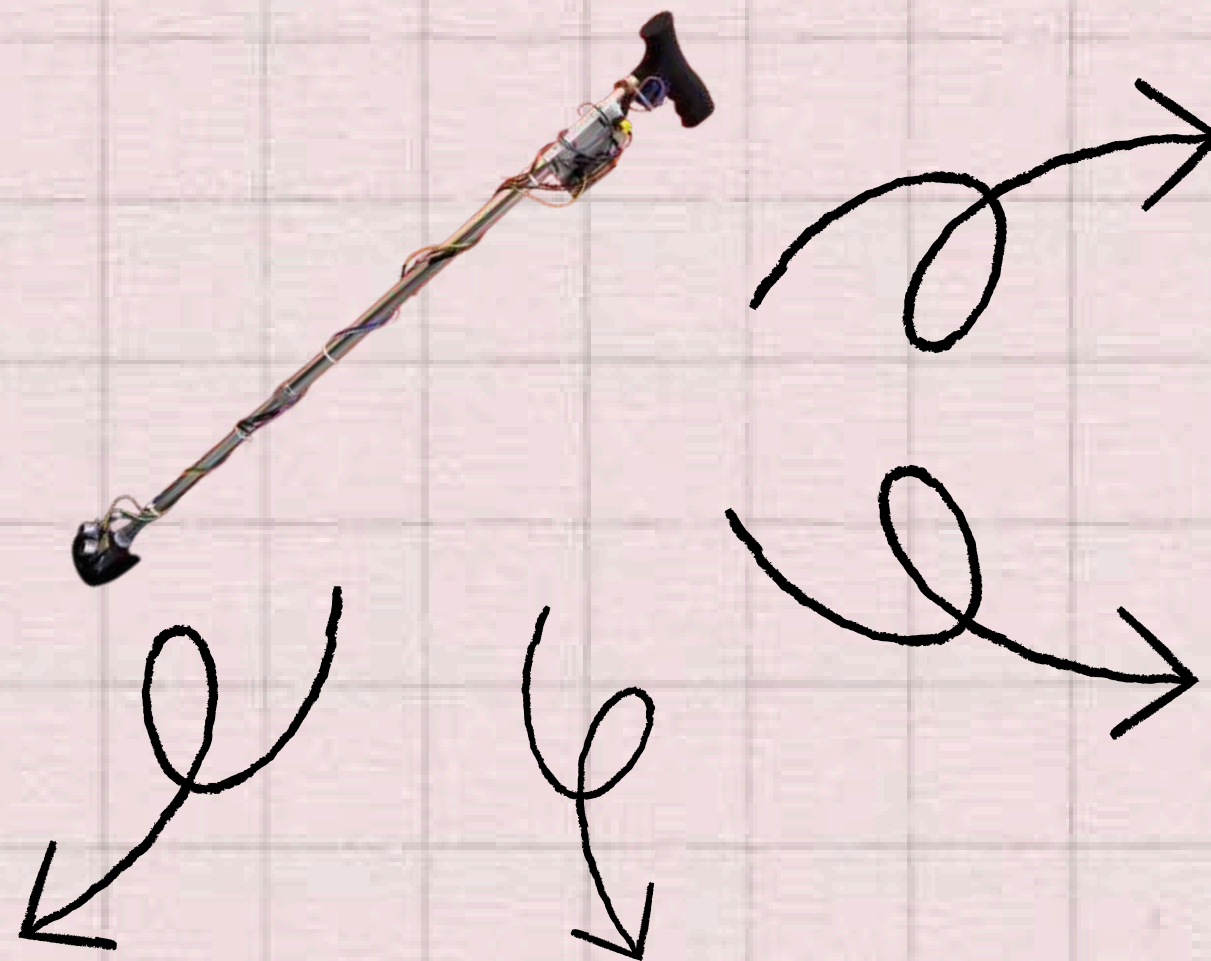
**alert obstacles
with vibration**

**improve safety
and independency**

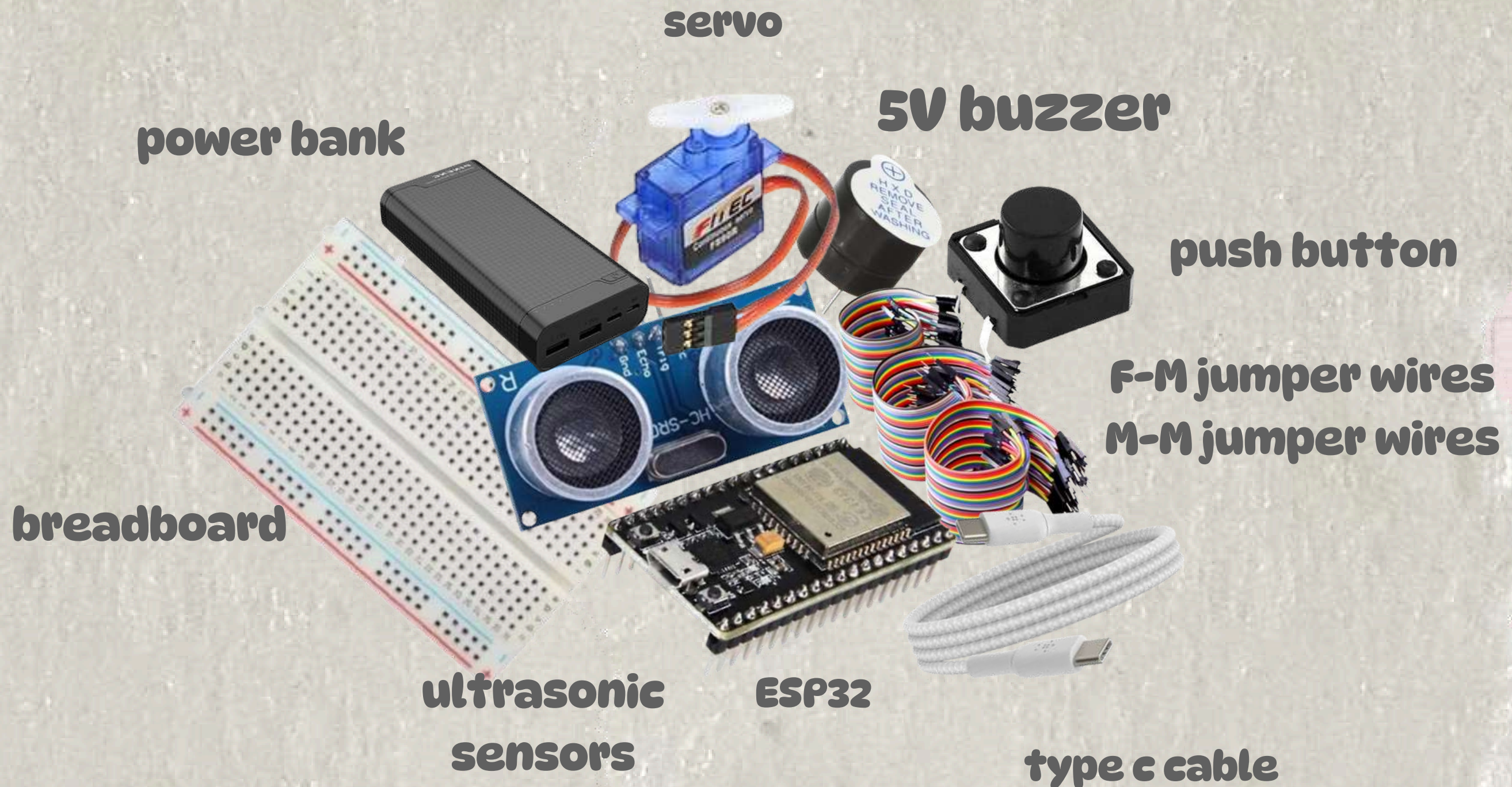
**produces sound
to alert user with
obstacles**

**servo produces motion to
alert user with obstacles**

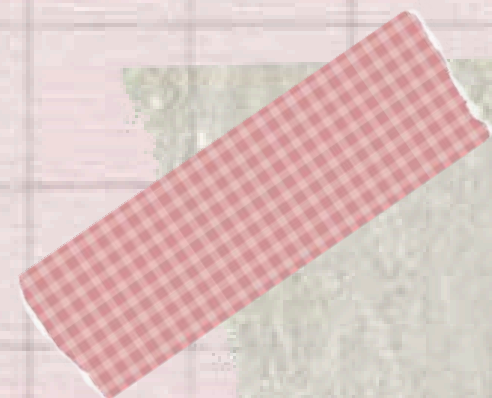
**alert family members
during user's SOS situation**



List of COMPONENTS



Lists of COMPONENTs



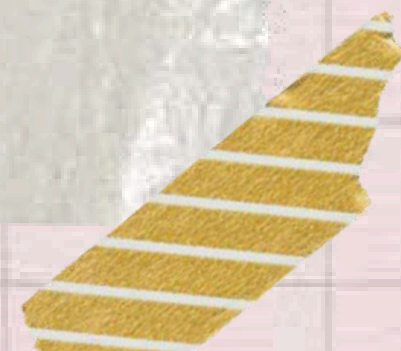
tape



cable tie



stick



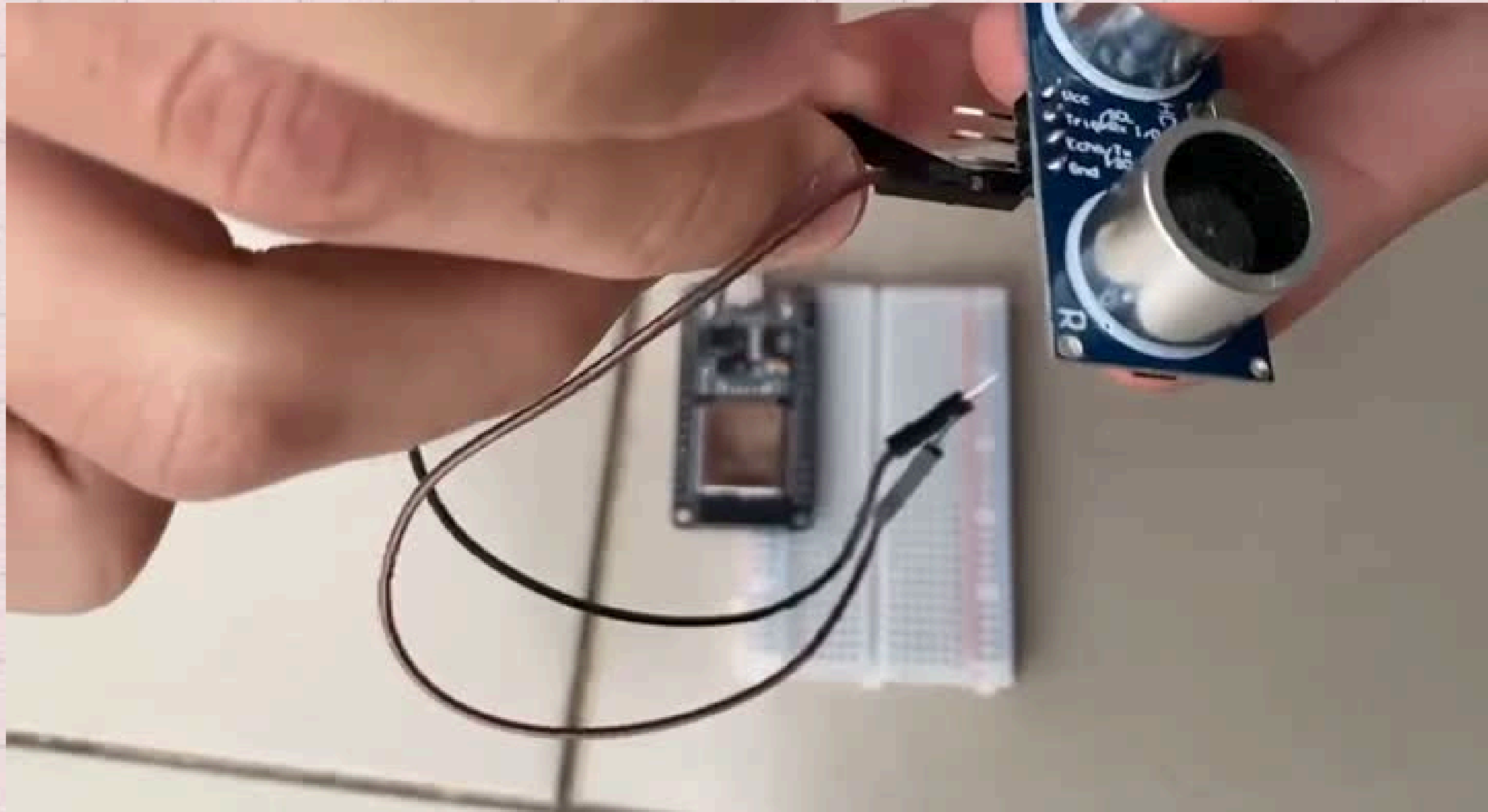
Coding

```
1 #include <ESP32Servo.h>
2 #include <WiFi.h>
3 #include <WiFiClientSecure.h>
4 #include <UniversalTelegramBot.h>
5 #include <ArduinoJson.h>
6
7 // ===== Pin Definitions =====
8 #define TRIG_PIN 12
9 #define ECHO_PIN 13
10 #define BUZZER_PIN 26
11 #define SERVO_PIN 32
12 #define BUTTON_PIN 22 // SOS button connected to GPIO22 → GND
13
14 // ===== Objects =====
15 Servo myServo;
16
17 // ===== WiFi Setup =====
18 const char* ssid = "stephen's";
19 const char* password = "ladi7777";
20
21 // ===== Telegram Setup =====
22 #define BOTtoken "8270811995:AAEOH7xiE6DBlqoJgUc7qEWstx6LJ-eWmQ8"
23
24 WiFiClientSecure client
25 UniversalTelegramBot bot(BOTtoken, client);
26
27 // ===== Multiuser Chat IDs =====
28 String chatIDs[] = {
29     "908355266",
30     "8127880120"
31 };
32 int totalUsers = sizeof(chatIDs) / sizeof(chatIDs[0]);
33
```

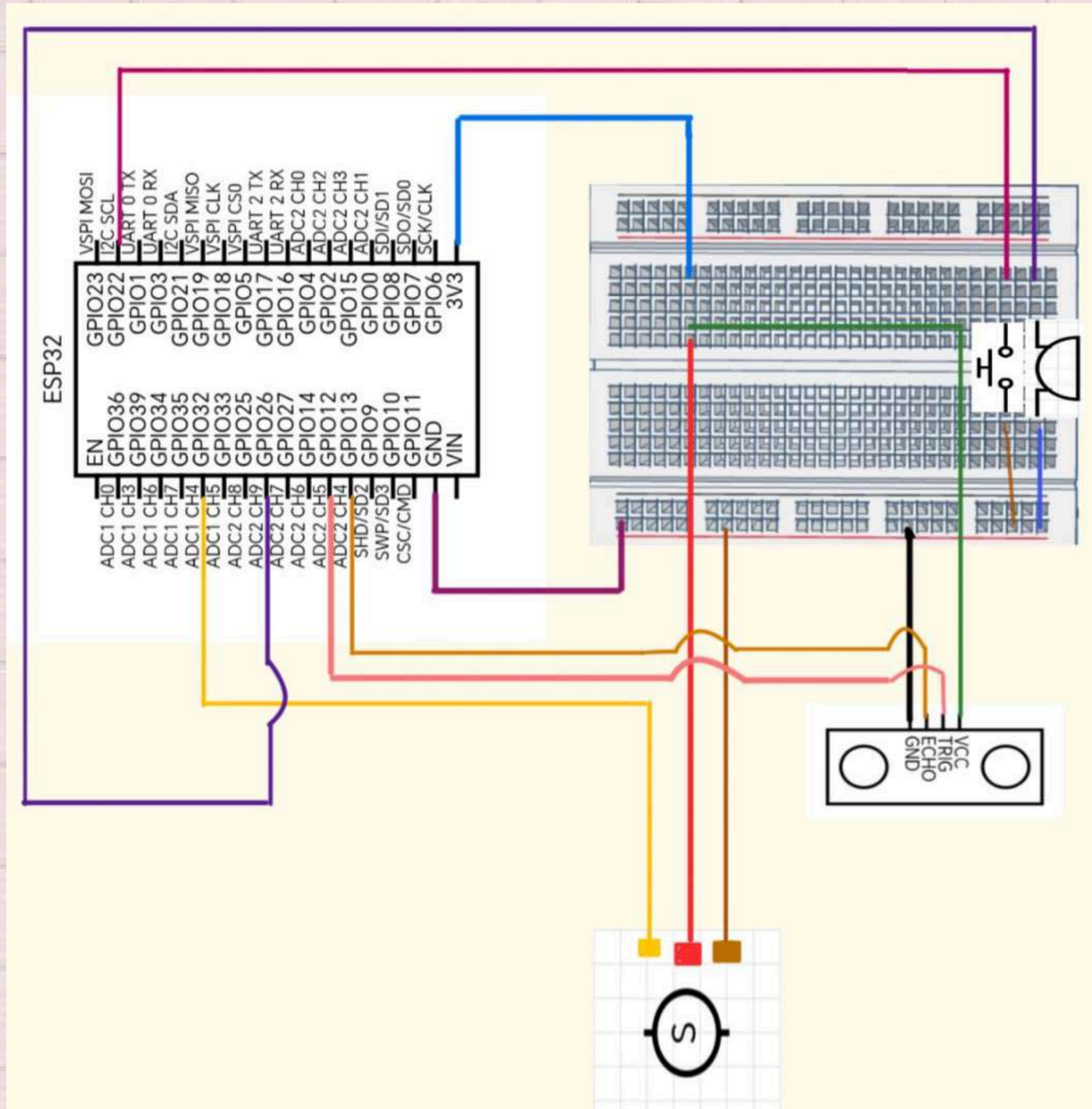
```
34 // ===== Helper Function to Send Message to All =====
35 void sendToAll(String message) {
36     for (int i = 0; i < totalUsers; i++) {
37         bot.sendMessage(chatIDs[i], message, "");
38     }
39 }
40
41 bool lastButtonState = HIGH;
42
43 void setup() {
44     Serial.begin(115200);
45
46     // Pin setup
47     pinMode(TRIG_PIN, OUTPUT);
48     pinMode(ECHO_PIN, INPUT);
49     pinMode(BUZZER_PIN, OUTPUT);
50     pinMode(BUTTON_PIN, INPUT_PULLUP); // Button uses internal pull-up
51
52     // Servo setup
53     myServo.attach(SERVO_PIN);
54     myServo.write(0); // Initial servo position
55
56     // WiFi setup
57     Serial.print("Connecting to WiFi");
58     WiFi.begin(ssid, password);
59     while (WiFi.status() != WL_CONNECTED) {
60         delay(500);
61         Serial.print(".");
62     }
63     Serial.println("\nConnected to WiFi!");
64
65     client.setInsecure(); // skip SSL certificate check
66
```

```
67 // Send Telegram boot-up message
68 sendToAll("✅ Smart Blind Stick is Online!");
69 }
70
71 void loop() {
72     // ===== SOS Button (Telegram) =====
73     bool buttonState = digitalRead(BUTTON_PIN);
74
75     if (buttonState == LOW && lastButtonState == HIGH) {
76         Serial.println("🚨 SOS Button Pressed!");
77         sendToAll("🚨 SOS Alert! Please help immediately!");
78         delay(1000); // debounce
79     }
80     lastButtonState = buttonState;
81
82     // ===== Ultrasonic Distance Measurement =====
83     digitalWrite(TRIG_PIN, LOW);
84     delayMicroseconds(2);
85     digitalWrite(TRIG_PIN, HIGH);
86     delayMicroseconds(10);
87     digitalWrite(TRIG_PIN, LOW);
88
89     long duration = pulseIn(ECHO_PIN, HIGH);
90     float distance = duration * 0.034 / 2;
91
92     Serial.print("Distance: ");
93     Serial.println(distance);
94
95     if (distance < 20) {
96         digitalWrite(BUZZER_PIN, HIGH); // Activate buzzer
97         myServo.write(90); // Move servo to 90°
98     } else {
99         digitalWrite(BUZZER_PIN, LOW); // Deactivate buzzer
100         myServo.write(0); // Reset servo to 0°
101     }
102
103     delay(200); // Small delay for stability
104 }
```

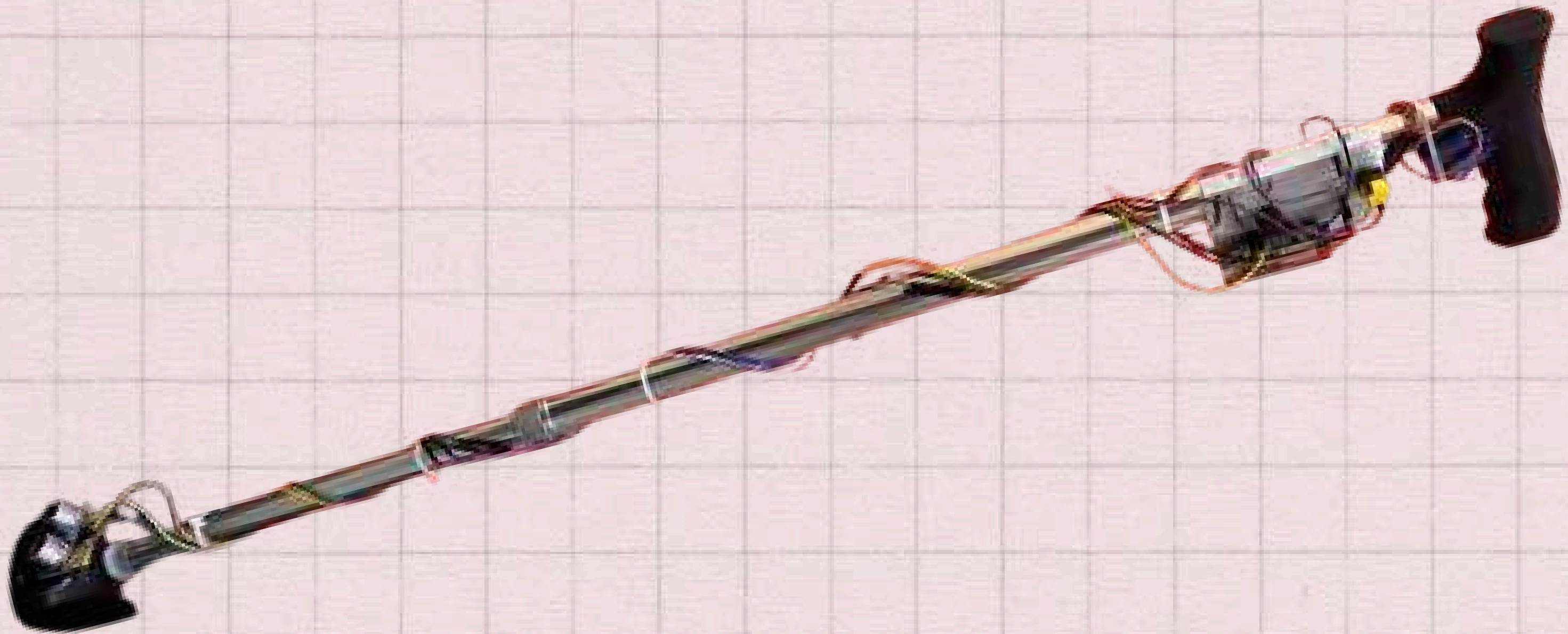
Circuit Assembly



schematic diagram



prototype





CONCLUSION

In summary, we achieved the aim to aid visually impaired individuals to ease their daily life by applying computing studies and technologies to solve problems. This system helps them to alert with obstacles within 50cm range by producing vibration and sound. Roughly, the cost of making one stick is cheap and affordable. Somehow, this stick can actually be improved with IoT which helps them with navigation and sends information to near family members about their whereabouts. Not to forget, this stick is also lightweight and can easily be carried anywhere. Statistically, this product potentially will help to reduce the numbers of injuries and deaths of visually impaired individuals.





REFERENCES

- **Dada Emmanuel Gbenga, Arhyel Ibrahim Shani and Adebimpe Lateef Adekunle, “Smart Walking Stick for Visually Impaired People Using Ultrasonic Sensors and Arduino.” International Journal of Engineering and Technology (IJET), vol 9, No. 5, pp. 3435 - 3447, 2017.**
 - **M. S. Nowak and J. Smigielski, “The Prevalence and Causes of Visual Impairment and Blindness among Older Adults in the City of Lodz, Poland.” Medicine, vol 94, number 5, pp. e505, February 2015 doi:10.1097/MD.0000000000000505**
 - **G. Gayathri, M. Vishnupriya, R. Nandhini and M. Banupriya “Smart Walking Stick for Visually Impaired.” International Journal of Engineering and Computer Science, vol. 3, number 3, pp. 4057-4061, 2014.**
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Thank you

